Universal Radiative Corrections for Precision Determinations of V_{ud}

BEACH conference

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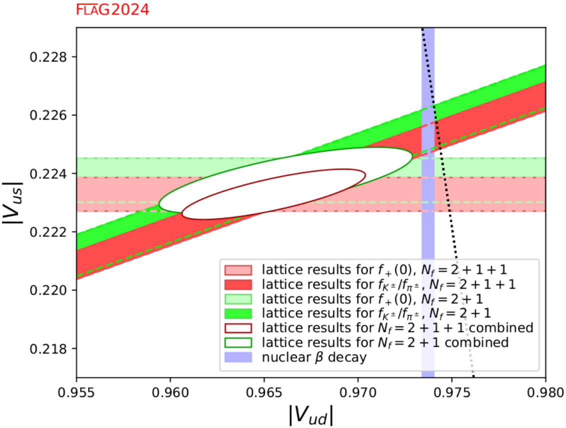
1. CKM matrix
2. V_{us} from kaon decays
3. V_{ud} EFT framework
4. V_{ud} from neutron decay
5. V_{ud} from pion β decay
6. NLO EW corrections
7. NNLO EW corrections

- **CKM matrix** $\longrightarrow$ quark flavour mixing in weak interactions;
- First-row unitarity test

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1,$$

  provides a sensitive probe of the SM;
- Current determinations of V_{ud} and V_{us} show a mild tension $\longrightarrow$ **Cabibbo Angle Anomaly**;
- Precision measurements of weak decays and accurate theoretical inputs are needed to clarify this tension;
- This talk:
  - Status of V_{us} from kaon decays ($K_{\ell 3}$ and $K_{\ell 2}$);
  - Status of V_{ud} from superallowed β decays, pion decay and neutron decay;
  - Ongoing work on **NNLO EW corrections for neutron decay** and V_{ud} extraction



V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays, FLAG2024.

FLAG2024, Seng(2502.04721)

$K_{\ell 3}$ **decays:**

$$K \to \pi \ell \nu_\ell, \qquad \ell = e, \mu,$$

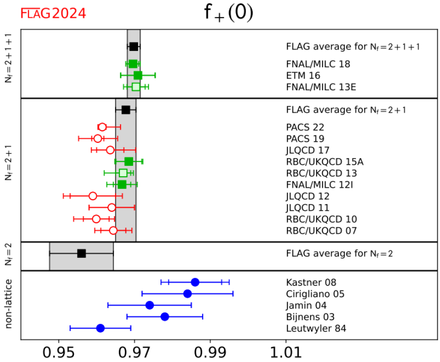
$$\Gamma(K_{\ell 3}) = \frac{G_F^2 M_K^5}{192\pi^3} C_K^2 \, S_{\rm EW} \, |V_{us}|^2 \left| f_+^{K^0\pi^-}(0) \right|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{\rm EM}^{K\ell}\right)^2$$

- $S_{\rm EW} = 1.0232(3)$: process-independent electroweak correction;
- $I_{K\ell}^{(0)}$: phase-space integral in the isospin limit;
- $\delta_{SU(2)}^{K\pi}$: strong isospin-breaking correction;
- $\delta_{\rm EM}^{K\ell}$: long-distance electromagnetic correction.

$$\langle \pi(p_\pi) | \bar{s}\gamma^\mu u | K(p_K) \rangle = (p_K + p_\pi)^\mu f_+(q^2) + (p_K - p_\pi)^\mu f_-(q^2),$$

Experimental input: $|V_{us}| \, f_+(0)$

Hadronic input: $f_+(0)$ from the vector form factor $\langle \pi | \bar{s}\gamma^\mu u | K \rangle$



Lattice determinations of $f_+(0)$. Black squares and grey bands denote FLAG averages.

Experimental input:

$$|V_{us}|f_+(0) = 0.21654(41)$$

FLAG average:

$$f_+(0) = 0.9698(17)$$

$$\Downarrow$$

$$|V_{us}| = 0.22328(58)$$

Ratio of V_{us}/V_{ud}

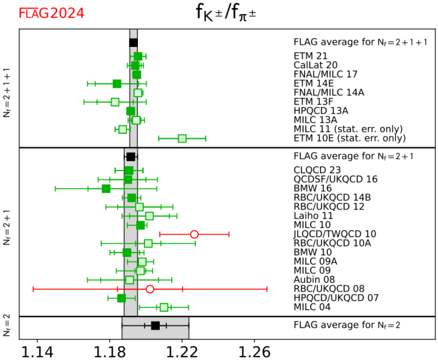
$$K_{\mu 2(\gamma)}: \qquad K^+ \to \mu^+ \nu_\mu(\gamma)$$

$$\frac{\Gamma(K^\pm_{\mu 2(\gamma)})}{\Gamma(\pi^\pm_{\mu 2(\gamma)})} = \left|\frac{V_{us}}{V_{ud}}\right|^2 \left(\frac{f_{K^\pm}}{f_{\pi^\pm}}\right)^2 \frac{M_{K^\pm}\left(1 - m_\mu^2/M_{K^\pm}^2\right)^2}{M_{\pi^\pm}\left(1 - m_\mu^2/M_{\pi^\pm}^2\right)^2}(1 + \delta_{\mathrm{EM}})$$

$$\langle 0|\bar{q}_1 \gamma^\mu \gamma_5 q_2|P(p)\rangle = i\, p^\mu f_P, \qquad P = \pi^+,\, K^+.$$

- Hadronic input encoded in the decay-constant ratio $f_{K^\pm}/f_{\pi^\pm}$;
- δ_{EM}: electromagnetic radiative correction;
- Lattice-QCD isospin corrections are systematically improvable;
- Isospin-breaking effects are relevant at current CKM precision.

Lattice determination of $f_{K^\pm}/f_{\pi^\pm}$



Lattice determinations of $f_{K^\pm}/f_{\pi^\pm}$. Black squares and grey bands denote FLAG averages.

Experimental input:

$$\left|\frac{V_{us}}{V_{ud}}\right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41)$$

FLAG average:

$$\frac{f_{K^\pm}}{f_{\pi^\pm}} = 1.1934(19)$$

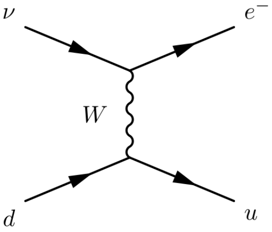
$$\Downarrow$$

$$\left|\frac{V_{us}}{V_{ud}}\right| = 0.23126(50)$$

$$(K_{\ell 2}/\pi_{\ell 2})$$

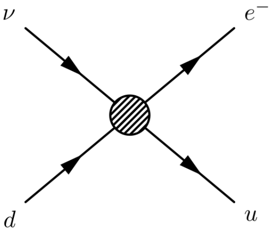
V_{ud} in the EFT framework

$$d \to u\, e\, \bar{\nu}_e$$



$$m_W \gg k$$

$$\Longrightarrow$$



Standard Model

Effective theory

$$\mathcal{H}_{\text{eff}} = \frac{G_F}{\sqrt{2}} V_{ud}\, C_q(\mu)\, Q_q$$

V_{ud} from neutron decay

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f(1 + 3g_A^2)(1 + \Delta_R^V)$$

- f phase-space factor
- g_A axial coupling
- Δ_R^V universal electroweak radiative correction.

$$\Delta_R^V = 2.436(16)\%$$

$$|V_{ud}|_n = 0.97410(41)$$

- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections;
- Reduced scale and scheme dependence;
- Improved CKM first-row unitarity tests;

Gorbahn, Jäger, Moretti (2025)

V_{ud} from pion β decay

$$\Gamma_{\pi\beta} = \frac{G_F^2 |V_{ud}|^2 M_\pi^5}{64\pi^3} |f_+^\pi(0)|^2 I_{\pi\ell}(1 + \Delta_{RC}^{\pi\ell})$$

- $f_+^\pi(0)$ pion vector form factor
- $I_{\pi\ell}$ phase-space integral
- $\Delta_{RC}^{\pi\ell}$ electroweak radiative correction (short distance and long distance)

$$\Delta_{RC}^{\pi\ell} = 0.03403(11)$$

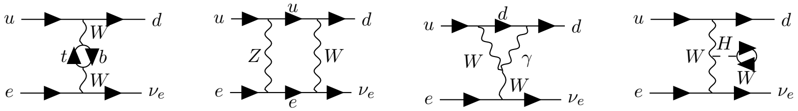
$$V_{ud}^\pi = 0.97346(283)$$

- NLL matching of LEFT and ChPT
- Theory uncertainty reduced by a factor ~ 3 (preparing theory for PIONEER)

Cirigliano, Hoferichter, Valori (2026)

Matching condition

$$\mathcal{A}_{\mathrm{SM}} = \mathcal{A}_{\mathrm{EFT}} = V_{ud} C_q Q_q$$



Representative one-loop EW diagrams contributing to the semi-leptonic process.

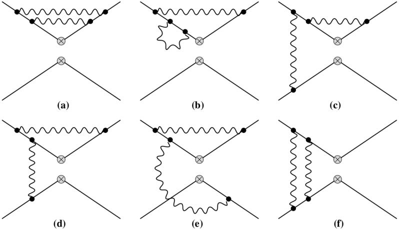
Wilson coefficient

$$C_q(\mu) = 1 + \frac{\alpha}{4\pi} C_q^{\mathrm{NLO}} + (\frac{\alpha}{4\pi})^2 C_q^{\mathrm{NNLO}}$$

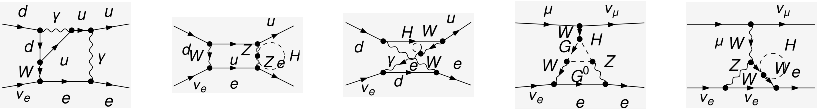
$$C_q^{\mathrm{NLO}} = -\frac{10}{3} - 2\ln\left(\frac{\mu^2}{M_Z^2}\right)$$

- Matching performed at the electroweak scale.
- Universal corrections cancel in the ratio of the quark decay normalized to muon decay: $\frac{C_q}{C_\mu}$.
- Only box-diagram contributions survive.

NNLO EW corrections



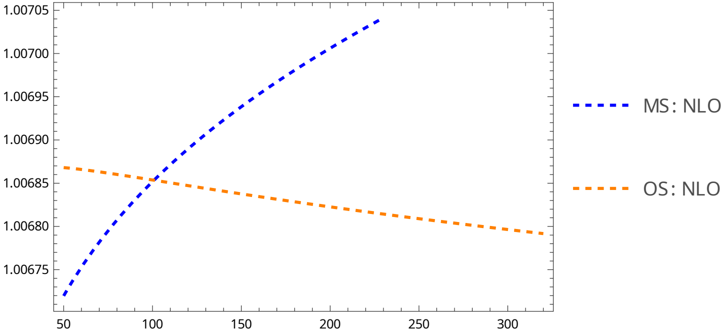
EFT NNLO EW corrections.



SM NNLO EW corrections.

- NNLO requires inclusion of boxes and penguins
- Tadpoles are included in the matching: gauge independent NNLO EW corrections and potentially substancial contribution in scheme conversion: $\overline{MS}$ to On-Shell.

Wilson-Coefficient Evolution



Scale dependence of the Wilson coefficient $C(2\,\mathrm{GeV})$ at NLO.

- Residual scale dependence at NLO induces an uncertainty $\delta C(2\,\mathrm{GeV}) \simeq \pm 2 \times 10^{-4}$, which is significant for precision determinations of V_{ud}.
- NNLO corrections reduce the uncertainty to $\delta C(2\,\mathrm{GeV}) < 2 \times 10^{-5}$, making perturbative uncertainties essentially negligible.

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.